Abhishek Yadav

Backend & Distributed Systems Engineer | Java 17 & Microservices

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PROFESSIONAL SUMMARY

Software Development Engineer specializing in distributed systems, high-concurrency architectures, and low-latency data storage using Java, Python, FastAPI, and Spring Boot. Passionate about building robust APIs, optimizing system throughput, and automating production workflows with clean code and rigorous unit testing.

WORK EXPERIENCE

Software Development Engineer (SDE) at Anantasutra May 2026 – Nov 2026

- Architected and deployed high-throughput Python data pipelines and RESTful microservices using **BeautifulSoup, Requests, and FastAPI**, boosting data ingestion by **40%** with automated verification and deduplication.
- Engineered automated **GitHub Actions CI/CD** pipelines and comprehensive unit test suites, reducing invalid transaction records by **30%** and ensuring zero-downtime microservice deployments.
- Implemented resilient error handling and asynchronous queueing with Celery and Redis, improving backend endpoint throughput under peak traffic loads.

Data Engineer Intern at Anantasutra Nov 2025 – Jan 2026

- Built automated lead-generation and web scraping pipelines with Python and RESTful API integrations, increasing data acquisition by 40% while improving data quality by 35% through backend verification.
- Developed microservices-based automation with GitHub Actions CI/CD and unit testing, improving overall data pipeline reliability and fault tolerance.

TECHNICAL PROJECTS

Distributed Key-Value Store (Java 17) **github.com/Abhiboss07/Distributed-KV-Store**

- Architected a high-concurrency in-memory KV store in Java 17 featuring a multi-client TCP server (FixedThreadPool) and O(1) LRU cache eviction.
- Implemented dual-strategy TTL expiry and AOF write-ahead logging with fsync policies and atomic compaction; verified via 144 unit and concurrency tests.

Voice AI SaaS Platform — Full-Stack Agentic Telephony **github.com/Abhiboss07/Voice-Agent**

- Built a cloud-native Voice AI SaaS platform with microservices backend (FastAPI, Redis, Celery, Docker), LiveKit SIP telephony, and OpenAI/Gemini voice models.
- Engineered WebSocket real-time metrics streaming, JWT authorization, and responsive React/TypeScript UI for low-latency interactive audio sessions.

Nexus Control Center — Linux System Telemetry & Diagnostics **github.com/Abhiboss07/Nexus**

- Developed an open-source cross-platform system telemetry and diagnostics control center using Rust, Tauri v2, React, and TypeScript.
- Designed modular plugin architecture with real-time process monitoring, memory analysis, and AI-assisted performance diagnostics.

TECHNICAL SKILLS

Languages: Java 17, Python, TypeScript, SQL, Rust, C++

Backend & Frameworks: FastAPI, Spring Boot, RESTful APIs, Microservices, gRPC

Cloud, CI/CD & DevOps: Docker, CI/CD, GitHub Actions, Linux, Kubernetes Basics

Distributed Systems & Storage: Redis, MongoDB, MySQL, PostgreSQL, Celery

APIs, Security & Protocols: RESTful API Design, WebSockets, JWT, RBAC, OAuth2

Testing & Reliability: Unit & Concurrency Testing (JUnit 5, PyTest), TDD

AI & Agentic Systems: OpenAI, Gemini, Agentic Workflows, Prompt Engineering

Frontend & UI: React, TypeScript, Tailwind CSS, HTML5/CSS3

Data Engineering & ETL: Data Pipelines, Web Scraping, Process Automation, ETL

Developer Tools: Git, VS Code, Cursor, Postman, Linux (Arch/CachyOS)

EDUCATION & COURSEWORK

Bachelor of Technology in Computer Science & Engineering 2023 – 2027 (Expected)

Veer Bahadur Singh Purvanchal University

Relevant Coursework: Distributed Systems, Object-Oriented Programming (Java), Data Structures & Algorithms, Computer Networks, Operating Systems, Database Management Systems

CERTIFICATIONS & HONORS

- **Deloitte Australia Technology Simulation:** Software Architecture, Cloud Infrastructure, Telemetry & Code Review (Forage).
- **CodSoft & Cognifyz Technologies:** Verified Java Software Development Certification, Advanced Object-Oriented Design.
- **Innovation Award:** Selected Participant & Pitch Presenter at Bharat Innovation Conclave.